

PPA-2: Reconfigurable Spatial Computational Substrate

Title

Reconfigurable Spatial Computing Using Symmetry-Governed Volumetric Logic Points

Field of the Invention

The invention relates to computing architectures and information processing systems, and more particularly to computational substrates in which computation occurs through spatial reconfiguration of discrete volumetric logic units governed by symmetry rather than fixed wiring or coordinates.

Background

Conventional computing substrates assume:

- fixed adjacency between logic elements,
- static wiring,
- global coordinate systems,
- and time-sequenced clocked execution.

Such assumptions severely limit scalability, robustness, and applicability to physical systems such as biological, chemical, or self-organizing materials, where adjacency and interaction networks are dynamic.

There is a need for a computational substrate in which:

- logic elements may dynamically rearrange neighbors,
 - computation arises from constrained spatial reconfiguration,
 - and global consistency is maintained without centralized control.
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Summary

The invention discloses a **reconfigurable spatial computational substrate** composed of discrete volumetric logic points, each having a finite internal state space governed by symmetry.

Computation proceeds through a two-phase process:

1. **Local state transitions** within each volumetric logic point, constrained by symmetry.
2. **Global spatial rearrangement** of adjacency relationships among logic points to maintain or restore symmetry consistency across the substrate.

Adjacency is not fixed. Neighbor relationships may change as a function of internal state and global symmetry requirements, allowing space itself to compute.

Detailed Description

Volumetric Logic Points

Each volumetric logic point is a discrete unit of space with:

- a finite internal state space,
- symmetry-constrained transitions,
- and no fixed positional identity.

The logic point is not defined by coordinates but by relational consistency with neighboring points.

Dynamic Adjacency

Unlike lattice-based systems (e.g., cubic grids), adjacency relationships are:

- mutable,
- symmetry-dependent,
- and globally constrained.

Neighbor rearrangement is computationally meaningful and part of the logic itself.

Two-Phase Computation Cycle

1. **Local Update Phase**

Each logic point updates its internal state using a symmetry-constrained gate.

2. **Global Reconciliation Phase**

Neighbor relationships are reconfigured to satisfy global symmetry constraints.

This cycle may repeat continuously or asynchronously.

Advantages

- Eliminates fixed wiring
 - No global clock required
 - Fault-tolerant by design
 - Naturally parallel
 - Applicable to physical and biological substrates
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Provisional Statement

This application claims priority for spatial computing substrates in which computation arises from symmetry-governed reconfiguration of volumetric logic points.